

Intrusion Detection in IoT Environments: An Experimental Evaluation Using Multiple ToN-IoT Dataset Subsets as a Baseline for Federated Learning

Rafael Sebastião Galdino¹ 

¹Department of Computer Science (Federal University of Lavras - UFLA)
P.O. Box 3037 – 37200-900 – Lavras – MG – Brazil

raffaelgaldino@gmail.com

Abstract. *The growing adoption of Internet of Things (IoT) devices has increased the need for efficient intrusion detection systems capable of identifying cyberattacks in heterogeneous environments. This work investigates the application of machine learning techniques for intrusion detection using data from the ToN-IoT dataset. Experiments were conducted using Logistic Regression, Random Forest and Multilayer Perceptron (MLP) models applied to different IoT devices, including Weather, Modbus, Garage Door and GPS Tracker. The proposed methodology includes data preprocessing, binary classification of normal and malicious traffic, and comparative evaluation using accuracy, precision, recall and F1-score metrics. The study establishes an experimental baseline for future integration with Federated Learning architectures in IoT environments. Preliminary results demonstrate significant performance differences among the evaluated models, highlighting the importance of model selection for intrusion detection tasks in IoT scenarios.*

1. Introduction

The Internet of Things (IoT) has promoted an increasing integration among connected devices, sensors, embedded systems, and distributed services, enabling applications in areas such as smart cities, healthcare, transportation, industry, and home automation. The expansion of these connected environments has brought significant advances in automation, monitoring, and real-time communication. However, the growing number of connected devices has also substantially increased the attack surface of networks, making IoT environments frequent targets of cyberattacks, such as Distributed Denial of Service (DDoS), botnets, vulnerability exploitation, and unauthorized access.

Unlike traditional computing environments, IoT devices often face limitations related to processing power, memory, storage capacity, and energy consumption, making the adoption of conventional security mechanisms particularly challenging. Furthermore, the heterogeneity of devices, protocols, and communication standards complicates the implementation of robust monitoring and protection solutions. In this context, Intrusion Detection Systems (IDS) have received increasing attention as mechanisms capable of identifying malicious behaviors and anomalous activities in real time.

In recent years, Machine Learning (ML) techniques have played a significant role in the development of intelligent IDS solutions for IoT environments, mainly due to their

ability to identify complex patterns and detect previously unknown attacks. Supervised, unsupervised, and deep learning-based approaches have been extensively investigated to improve detection capabilities and reduce false positives in distributed and heterogeneous environments. Recent studies have also highlighted Federated Learning (FL) as a promising alternative for distributed training of detection models without requiring data centralization, contributing to enhanced privacy, scalability, and reduced network traffic in Edge Computing environments.

Despite recent advances, important challenges remain regarding the computational cost of models, the resource constraints of IoT devices, and the feasibility of deploying intelligent solutions in edge environments. Moreover, many existing studies focus on complex Deep Learning architectures that require substantial computational resources, limiting their practical applicability in resource-constrained devices.

Given this scenario, this work investigates the application of Machine Learning techniques for intrusion detection in IoT environments using subsets of the ToN-IoT dataset. The study evaluates the performance of different supervised classification models, including Logistic Regression, Random Forest, and Multilayer Perceptron (MLP), considering performance metrics widely adopted in the literature, such as Accuracy, Precision, Recall, and F1-Score.

The experiments employ different subsets of the ToN-IoT dataset, including Weather, Modbus, Garage Door, and GPS Tracker devices, enabling the analysis of model behavior across heterogeneous Internet of Things scenarios. The obtained results establish an experimental baseline for future evaluations involving distributed architectures based on Federated Learning.

As a continuation of this research, a federated architecture based on the Federated Averaging (FedAvg) algorithm will be implemented, enabling the investigation of the impact of distributed training on intrusion detection in IoT and Edge Computing environments. In this way, the study seeks to contribute to the development of more efficient, scalable IDS solutions aligned with the privacy requirements of modern Internet of Things ecosystems.

The main contributions of this work are: (i) a comparative evaluation of different supervised Machine Learning algorithms applied to intrusion detection across multiple IoT devices from the ToN-IoT dataset; (ii) an analysis of the impact of device-specific characteristics on classifier performance; and (iii) the establishment of an experimental baseline for future Federated Learning-based implementations in distributed IoT environments.

2. Theoretical Background and Related Work

The Internet of Things (IoT) comprises an ecosystem of connected devices capable of continuously collecting, processing, and sharing data over the Internet. Sensors, actuators, cameras, mobile devices, and industrial equipment are integrated into this environment, enabling applications in smart cities, industrial automation, healthcare, agriculture, and intelligent transportation. Despite the benefits associated with connectivity and automation, the expansion of IoT has also significantly increased the attack surface of networks, as devices often exhibit limitations in processing power, storage capacity, and embedded security mechanisms [Bharati and Podder 2022, Alfahaid et al. 2025].

In this context, attacks such as Distributed Denial of Service (DDoS), firmware vulnerability exploitation, botnets, and unauthorized access have become increasingly common in IoT environments. Furthermore, the heterogeneity of devices and communication protocols complicates the implementation of traditional security solutions, making it necessary to adopt intelligent and adaptive mechanisms for network monitoring and protection [Haji and Ameen 2021, Rahman et al. 2025].

2.1. Intrusion Detection Systems in IoT

Intrusion Detection Systems (IDS) are designed to identify suspicious or malicious activities in networks and computing devices, serving as a complement to traditional security mechanisms such as firewalls and access control lists. In general, IDS can be classified into two main approaches: signature-based systems and anomaly-based systems [Gassais et al. 2020, Haji and Ameen 2021].

Signature-based IDS rely on previously known attack patterns to identify malicious activities. Although they provide high accuracy for known threats, they have limitations in detecting novel attacks or previously unseen variants. In contrast, anomaly-based IDS build models of normal network behavior and identify significant deviations as potential intrusions, making them more suitable for dynamic and heterogeneous environments such as IoT networks [Malathi and Padmaja 2023].

The development of IDS for IoT environments presents additional challenges associated with the computational constraints of devices, including limitations in memory, processing power, and energy consumption. Furthermore, real-time applications require low latency and rapid response capabilities to prevent the large-scale propagation of attacks. Consequently, lightweight and scalable solutions have been extensively investigated in the literature, particularly those supported by Machine Learning techniques and Edge Computing paradigms [Kaushik et al. 2025, Rahman et al. 2025].

2.2. Machine Learning Applied to IDS in IoT

Machine Learning (ML) techniques have been widely employed to enhance Intrusion Detection Systems in IoT environments due to their ability to identify complex patterns and automatically detect anomalous behaviors [Bharati and Podder 2022, Kaushik et al. 2025]. In general, these techniques can be categorized into supervised methods, unsupervised methods, and deep learning-based approaches.

In supervised methods, models are trained using previously labeled data, learning patterns associated with legitimate and malicious traffic. Algorithms such as Decision Trees and Random Forest stand out for their high classification performance, interpretability, and relatively low computational cost, characteristics that are particularly important for applications deployed on edge devices and resource-constrained environments [Kaushik et al. 2025, Alomiri et al. 2024].

On the other hand, unsupervised methods, such as K-means and DBSCAN, do not rely on labeled data during training. These approaches aim to identify natural groupings within the data or detect instances considered anomalous with respect to the prevailing network behavior. In IoT scenarios, such methods are especially relevant due to the difficulty of obtaining large volumes of labeled data and the continuous evolution of attack patterns [Haji and Ameen 2021, Zhang and Liu 2022].

In addition to traditional ML approaches, Deep Learning (DL) techniques have also been explored for IoT-based IDS. Recurrent neural networks, autoencoders, and hybrid architectures have demonstrated high performance in detecting complex attacks and malicious traffic. Recent studies have combined deep learning with technologies such as blockchain and Federated Learning to enhance security while preserving the privacy of distributed data [Alsharif et al. 2023, Bhavsar et al. 2024, Vargas et al. 2021]. However, these approaches often require greater computational resources, which may limit their deployment on resource-constrained edge devices.

2.3. Federated Learning Applied to IDS in IoT

Federated Learning (FL) has emerged as a promising approach for distributed training of Machine Learning models without requiring data centralization. In this paradigm, local devices perform training using their own data and share only the updated model parameters with a central server, reducing risks associated with privacy breaches and the exposure of sensitive information [McMahan et al. 2017].

In IoT environments, FL offers important advantages due to the distributed nature of networks and the growing concern regarding data privacy and security. In addition to reducing the transmission of sensitive information to centralized servers, Federated Learning enables collaborative training among heterogeneous and geographically distributed devices [Kairouz et al. 2021, Li et al. 2020].

Several recent studies have investigated the application of FL to IDS for IoT environments. Bhavsar et al. proposed FL-IDS, a Federated Learning-based intrusion detection system for Transportation IoT environments, demonstrating improvements in both privacy preservation and distributed performance [Bhavsar et al. 2024]. Other studies have explored hybrid models combining deep neural networks and FL to enhance attack detection capabilities in decentralized environments [Zhang et al. 2024, Sharma et al. 2025].

Despite recent advances, significant challenges remain regarding data heterogeneity across clients, communication overhead, device synchronization, and computational constraints in edge environments. Consequently, current research efforts focus on developing lighter, more efficient, and more robust models to support the practical deployment of FL in real-world IoT scenarios [Onsu et al. 2023, Alghamdi et al. 2025].

2.4. Datasets for IDS in IoT

The evaluation of Machine Learning-based IDS relies directly on the availability of representative datasets. Among the most widely used datasets in the literature are NSL-KDD, UNSW-NB15, CICIDS2017, and ToN-IoT [Moustafa et al. 2020, Rahman et al. 2025].

NSL-KDD is an evolution of the traditional KDD Cup 1999 dataset and remains widely used due to its reduced redundancy and improved balance between normal and malicious classes. Although it does not fully represent contemporary traffic in IoT environments, the dataset continues to be relevant for the initial evaluation of intrusion detection algorithms [Rahman et al. 2025].

UNSW-NB15 and CICIDS2017, in turn, incorporate more modern attack scenarios and traffic characteristics that are closer to those observed in contemporary real-world

networks. The ToN-IoT dataset stands out specifically for encompassing IoT and Industrial IoT environments, including sensor data, telemetry information, and various types of distributed attacks, making it an important benchmark in recent research on IDS and Federated Learning [Moustafa et al. 2020, Gad et al. 2022].

The use of multiple datasets enables the evaluation not only of the predictive performance of models but also of their robustness and generalization capability across heterogeneous scenarios [Kaushik et al. 2025, Rahman et al. 2025].

2.5. Synthesis of Related Work and Research Gap

Recent studies on intrusion detection in IoT environments have demonstrated significant advances in the use of Machine Learning, Deep Learning, and Federated Learning for the identification of cyberattacks. Supervised models frequently achieve high Accuracy in scenarios with labeled data, while unsupervised and deep learning-based approaches have been employed for anomaly detection and the identification of previously unknown threats [Haji and Ameen 2021, Malathi and Padmaja 2023, Kaushik et al. 2025].

Furthermore, recent research highlights a growing interest in the adoption of Federated Learning in IoT environments, primarily due to its advantages in terms of privacy preservation, decentralization, and edge-based processing [Bhavsar et al. 2024, Kairouz et al. 2021]. However, many existing studies still exhibit high computational complexity, dependence on robust infrastructure, or a limited focus on specific experimental scenarios.

In this context, the present work investigates the application of Machine Learning techniques to IDS for IoT environments while simultaneously considering predictive performance, computational cost, and deployment feasibility in edge environments. In addition, the study takes into account recent advances in the literature on Federated Learning and the challenges associated with distributed security in IoT ecosystems, contributing to the discussion of lighter, more scalable solutions suitable for resource-constrained devices. Although the present study initially adopts centralized Machine Learning approaches, the future incorporation of Federated Learning-based strategies represents a promising direction for enhancing privacy, scalability, and suitability for distributed IoT environments.

3. Methodology

This work investigates the application of Machine Learning techniques to Intrusion Detection Systems (IDS) for IoT environments, considering predictive performance, computational cost, and execution feasibility in Edge Computing environments. Furthermore, the study establishes a preliminary architecture based on Federated Learning (FL), aiming to support future distributed IDS applications in IoT networks.

The methodology was organized into sequential stages involving dataset selection, data preprocessing, model training, experimental evaluation, and validation in an edge computing environment.

3.1. Proposed Architecture

The architecture considered in this work is based on a distributed IoT environment composed of edge devices responsible for network traffic collection and analysis. Initially, the

models are trained and evaluated in a centralized manner, establishing an experimental baseline for performance comparison.

Subsequently, a Federated Learning-based architecture is proposed, in which multiple devices perform local training using their own data and share only the updated model parameters with a central server responsible for global aggregation. This approach reduces the need for direct data sharing, contributing to privacy preservation and reduced network traffic, as discussed in recent studies from the literature [McMahan et al. 2017, Kairouz et al. 2021, Li et al. 2020].

The proposed architecture follows the client-server paradigm commonly adopted in recent federated approaches for intrusion detection in IoT environments [Bhavsar et al. 2024, Alghamdi et al. 2025].

3.2. Datasets Used

For the experimental evaluation, the ToN-IoT (Telemetry of Things and Industrial Internet of Things) dataset was employed, as it has been widely used in recent research on intrusion detection in IoT and IIoT environments. The dataset includes data collected from real IoT devices, sensors, industrial systems, and multiple categories of cyberattacks, enabling the evaluation of model behavior across heterogeneous scenarios.

In this work, specific subsets of the ToN-IoT dataset were utilized, including the Weather, Modbus, Garage Door, and GPS Tracker devices. Each subset presents distinct traffic and telemetry characteristics, allowing the analysis of detection capabilities under different application contexts.

The Weather subset contains information related to temperature, atmospheric pressure, and humidity. The Modbus subset comprises records associated with the Modbus industrial protocol, which is widely used in industrial automation systems. The Garage Door and GPS Tracker subsets represent typical IoT devices found in residential and tracking environments, respectively.

The use of multiple devices enables the assessment of model robustness in the presence of the heterogeneity that characterizes modern IoT environments and establishes an experimental baseline for future evaluations involving Federated Learning-based architectures.

Table 1. Characteristics of the ToN-IoT Subsets Used

Subset	Instances	Attributes Used	Classes
Modbus	31.106	FC1, FC2, FC3, FC4	Normal/Attack
Weather	39.260	Temperature, Pressure, Humidity	Normal/Attack
Garage Door	39.587	Door State, Smart-phone Signal	Normal/Attack
GPS Tracker	38.960	Latitude, Longitude	Normal/Attack

3.3. Data Preprocessing

Initially, the selected subsets of the ToN-IoT dataset were loaded. Attributes related to date and time were removed from the experiments because they do not represent characteristics directly associated with attack behavior.

Table 2. Class Distribution Across the Evaluated Subsets

Subset	Normal	Attack
Modbus	15.000	16.106
Weather	15.000	24.260
Garage Door	15.000	24.587
GPS Tracker	15.000	23.960

For the classification experiments, a binary classification approach was adopted, using the *label* variable as the target attribute, where the value 0 represents normal traffic and the value 1 represents malicious traffic.

The data were divided into training and testing sets using an 80%/20% split, with class stratification applied to preserve the original class distribution.

For models requiring feature normalization, the StandardScaler technique was employed to standardize the data with zero mean and unit variance. This procedure was primarily applied to Logistic Regression and Artificial Neural Network-based models.

3.4. Machine Learning Models

Three supervised models widely used in classification tasks were evaluated: Logistic Regression, Random Forest, and Multilayer Perceptron (MLP).

Logistic Regression was employed as a baseline model due to its computational simplicity and widespread use in binary classification problems.

Random Forest was selected because of its ability to capture nonlinear relationships among features and its consistent performance in intrusion detection applications.

The MLP model was included as it represents an artificial neural network-based approach, which has been frequently adopted in recent studies involving IDS and Federated Learning applications.

All models were implemented using the Scikit-Learn library and evaluated under the same experimental conditions.

The experiments employed the default parameters provided by the Scikit-Learn library. Logistic Regression was configured using the *lbfgs* solver. Random Forest was configured with 100 decision trees. The MLP model was trained using a single hidden layer with 100 neurons and a maximum limit of 300 training epochs.

Table 3. Configuration of the Evaluated Models

Model	Configuration
Logistic Regression	solver = lbfgs
Random Forest	n_estimators = 100
MLP	hidden_layer_sizes = (100), max_iter = 300

3.5. Federated Learning

Although the primary focus of this stage of the study is the centralized evaluation of Machine Learning models, the experiments were designed to serve as a baseline for future implementations based on Federated Learning.

The proposed approach consists of distributing subsets of the ToN-IoT dataset among different simulated clients, representing distinct IoT devices. Each client performs local training using its own data and shares only the updated model parameters with a central server responsible for global aggregation.

For aggregating the local models, the Federated Averaging (FedAvg) strategy will be adopted, as it is one of the most widely used approaches in Federated Learning systems due to its simplicity and efficiency [McMahan et al. 2017].

This approach reduces the need for direct data sharing, thereby contributing to privacy preservation and reducing communication overhead among participating devices [Kairouz et al. 2021, Li et al. 2020].

Therefore, the results presented in this work represent an essential initial step toward future evaluations involving Federated Learning applied to intrusion detection in distributed IoT environments [Bhavsar et al. 2024].

3.6. Experimental Environment

The experiments were developed using the Python programming language and executed in a Jupyter Notebook environment.

The Pandas library was employed for data manipulation, NumPy for numerical operations, and Scikit-Learn for the implementation of Machine Learning algorithms and model evaluation.

The experiments were conducted in a conventional computing environment in order to validate the feasibility of the proposed methodology before implementing distributed scenarios based on Federated Learning.

4. Results

The experiments were conducted using four subsets of the ToN-IoT dataset: Modbus, Weather, Garage Door, and GPS Tracker. Three supervised Machine Learning algorithms were evaluated: Logistic Regression, Random Forest, and Multilayer Perceptron (MLP).

The models were trained and evaluated on binary classification tasks, considering normal traffic and malicious traffic. Performance was analyzed using the metrics Accuracy, Precision, Recall, and F1-Score. Table 4 summarizes the Accuracy results obtained by the evaluated algorithms across the different subsets of the ToN-IoT dataset.

Table 4. Accuracy of the Evaluated Models Across the ToN-IoT Subsets

Subset	Logistic Regression	MLP	Random Forest
Modbus	51,41%	57,65%	98,46%
Weather	60,93%	80,41%	98,47%
Garage Door	100,00%	100,00%	100,00%
GPS Tracker	100,00%	100,00%	100,00%

The experiments were conducted using four subsets of the ToN-IoT dataset: Modbus, Weather, Garage Door, and GPS Tracker. Three supervised Machine Learning algorithms were evaluated: Logistic Regression, Random Forest, and Multilayer Perceptron (MLP).

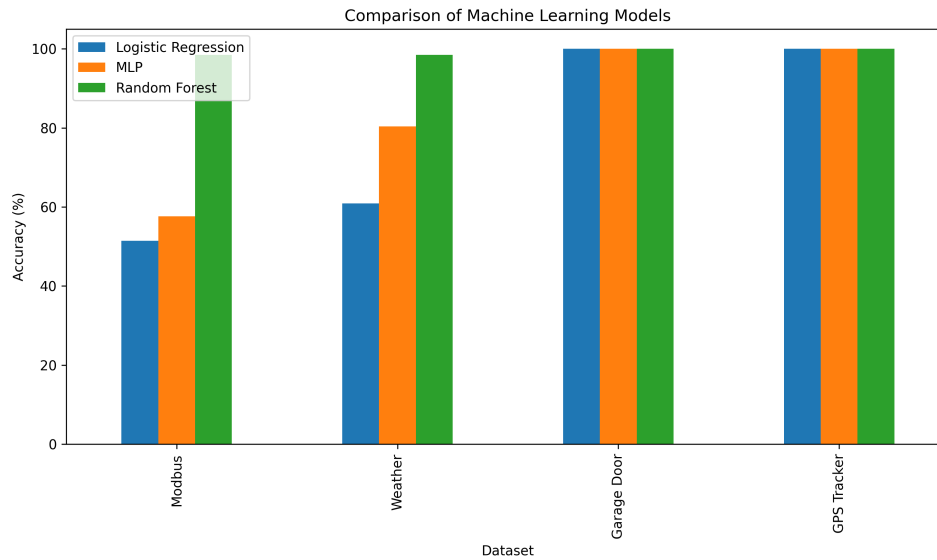


Figure 1. Comparison of the Accuracy of the Evaluated Models Across the ToN-IoT Subsets.

Table 5. Performance Metrics of Random Forest Across the Evaluated Subsets

Subset	Accuracy	Precision	Recall	F1-score
Modbus	98,46%	0,99	0,98	0,98
Weather	98,47%	0,99	0,99	0,99
Garage Door	100,00%	1,00	1,00	1,00
GPS Tracker	100,00%	1,00	1,00	1,00

The models were trained and evaluated on binary classification tasks, considering normal traffic and malicious traffic. Performance was analyzed using the metrics Accuracy, Precision, Recall, and F1-Score. Table 4 summarizes the Accuracy results obtained by the evaluated algorithms across the different subsets of the ToN-IoT dataset.

It can be observed that Random Forest achieved the best results across all evaluated scenarios, reaching Accuracy values above 98% for the Modbus and Weather subsets and achieving perfect performance on the Garage Door and GPS Tracker subsets.

Logistic Regression exhibited lower performance on the Modbus and Weather datasets, suggesting limitations in modeling the more complex patterns present in these devices. For example, in the Modbus subset, the model achieved an Accuracy of only 51.41%, whereas Random Forest reached 98.46%.

The MLP model achieved intermediate results. Although it outperformed Logistic Regression on the Weather subset, reaching an Accuracy of 80.41%, its performance remained below that achieved by Random Forest.

These results demonstrate that different IoT devices present distinct levels of complexity for intrusion detection, directly influencing the performance of the evaluated algorithms.

5. Discussion

The results obtained reveal significant differences among the evaluated ToN-IoT subsets.

While the Modbus and Weather devices presented more challenging classification scenarios, the Garage Door and GPS Tracker subsets exhibited a high degree of class separability, enabling all evaluated models to achieve 100% Accuracy. Exploratory data analysis indicated that the attributes of these devices display markedly distinct patterns between normal and malicious traffic, facilitating correct class identification by the evaluated algorithms.

For the Modbus subset, Random Forest substantially outperformed the other algorithms, achieving an Accuracy of 98.46%. This result suggests that the attack patterns associated with the Modbus protocol exhibit nonlinear characteristics that are more effectively captured by decision tree-based models.

The Weather subset exhibited intermediate behavior. Although Logistic Regression achieved an Accuracy of only 60.93%, the MLP model reached 80.41%, indicating that techniques capable of modeling nonlinear relationships are better suited to capturing the patterns present in environmental sensor data.

For the Garage Door and GPS Tracker subsets, all algorithms achieved perfect performance. Exploratory analysis revealed a high degree of separability between the normal and attack classes, substantially reducing the complexity of the classification task.

Overall, the experiments demonstrate that the selection of a Machine Learning model should take into account the specific characteristics of the monitored device and the dataset being analyzed. Among the evaluated algorithms, Random Forest exhibited the most consistent behavior, making it a strong candidate for future implementations in Federated Learning environments.

The experiments also highlight the importance of evaluating multiple devices and scenarios before deploying distributed architectures based on Federated Learning. Consequently, the results obtained in this stage provide an experimental baseline for future comparative studies.

6. Threats to Validity

The primary threats to the validity of this study are related to the use of only four subsets of the ToN-IoT dataset (Weather, Modbus, Garage Door, and GPS Tracker), the evaluation of a limited number of supervised algorithms, and the absence of experiments in a federated environment at this stage of the research.

Although the selected subsets represent different IoT devices and enable the evaluation of heterogeneous scenarios, other devices available within the ToN-IoT dataset or in alternative datasets may exhibit different characteristics, potentially affecting model performance.

Furthermore, the experiments were conducted in a centralized environment and therefore do not account for specific aspects of Federated Learning architectures, such as client heterogeneity, communication overhead, device synchronization, and the impact of distributed model aggregation.

Consequently, future studies involving a broader range of IoT devices, additional datasets, and Federated Learning-based implementations will be necessary to enhance the generalizability of the results obtained.

Another threat to validity is related to the class imbalance observed in some subsets of the ToN-IoT dataset, which may influence classifier performance and favor aggregate metrics such as Accuracy.

In addition, the experiments employed only a single train-test split. Techniques such as cross-validation may be adopted in future studies to reduce potential variations resulting from data sampling.

7. Conclusion

This work investigated the application of Machine Learning techniques for intrusion detection in IoT environments using subsets of the ToN-IoT dataset. Logistic Regression, Random Forest, and Multilayer Perceptron (MLP) models were evaluated in binary classification tasks involving normal and malicious traffic.

The experimental results revealed significant differences among the evaluated devices. While the Garage Door and GPS Tracker subsets exhibited high class separability, allowing all models to achieve 100% Accuracy, the Modbus and Weather subsets presented more challenging scenarios. In these cases, Random Forest achieved superior performance, reaching approximately 98% Accuracy and outperforming the other evaluated algorithms.

The obtained results established a consistent experimental baseline for future evaluations involving Federated Learning-based architectures. Furthermore, the findings demonstrated that different IoT devices may present distinct levels of complexity for intrusion detection, reinforcing the importance of selecting appropriate algorithms according to the characteristics of the monitored environment.

As future work, a federated architecture based on the Federated Averaging (FedAvg) algorithm will be implemented, distributing the training process across multiple simulated IoT devices. This will enable the evaluation of the impact of distributed training on intrusion detection performance and facilitate comparisons between centralized and federated models in Edge Computing environments.

Artifact Availability

The source code, experimental notebooks, results, and supplementary materials used in this study are available at:

GitHub Repository

References

- [Alfahaid et al. 2025] Alfahaid, A., Alalwany, E., Almars, A. M., Alharbi, F., Atlam, E., and Mahgoub, I. (2025). Machine learning-based security solutions for iot networks: A comprehensive survey. *Sensors*, 25(11).
- [Alghamdi et al. 2025] Alghamdi, A. et al. (2025). An optimal federated learning-based intrusion detection for iot environment. *Scientific Reports*, 15.

- [Alkali et al. 2022] Alkali, Y., Routray, I., and Whig, P. (2022). Study of various methods for reliable, efficient and secured iot using artificial intelligence. In *Proceedings of the International Conference on Innovative Computing & Communication (ICICC)*.
- [Alomiri et al. 2024] Alomiri, A., Mishra, S., and AlShehri, M. (2024). Machine learning-based security mechanism to detect and prevent cyber-attack in iot networks. *International Journal of Computing and Digital Systems*, 16(1):645–659.
- [Alsharif et al. 2023] Alsharif, N. A., Mishra, S., and Alshehri, M. (2023). Ids in iot using machine learning and blockchain. *Engineering, Technology & Applied Science Research*, 13(4):11197–11203.
- [Alyami et al. 2023] Alyami, H. et al. (2023). Deep learning and federated learning for iot security: A survey. *Electronics*, 12(9):2103.
- [Alzhrani and Alliheedi 2024] Alzhrani, R. and Alliheedi, M. (2024). Enhancing iot security in 5g networks: Mitigating ddos attacks with deep learning. *Journal of Information Security and Cybercrimes Research*, 7(2):156–166.
- [Bharati and Podder 2022] Bharati, S. and Podder, P. (2022). Machine and deep learning for iot security and privacy: applications, challenges, and future directions. *Security and communication networks*, 2022(1):8951961.
- [Bhavsar et al. 2024] Bhavsar, M. H., Bekele, Y. B., Roy, K., Kelly, J. C., and Limbrick, D. (2024). FI-ids: Federated learning-based intrusion detection system using edge devices for transportation iot. *IEEE Access*, 12:52215–52226.
- [Canadian Institute for Cybersecurity 2017] Canadian Institute for Cybersecurity (2017). Cicides2017 dataset. University of New Brunswick.
- [Gad et al. 2022] Gad, A. R., Haggag, M., Nashat, A. A., and Barakat, T. M. (2022). A distributed intrusion detection system using machine learning for iot based on ton-iot dataset. *International Journal of Advanced Computer Science and Applications*, 13(6).
- [Gassais et al. 2020] Gassais, R., Ezzati-Jivan, N., Fernandez, J. M., Aloise, D., and Dagenais, M. R. (2020). Multi-level host-based intrusion detection system for internet of things. *Journal of Cloud Computing*, 9(1):62.
- [Haji and Ameen 2021] Haji, S. H. and Ameen, S. Y. (2021). Attack and anomaly detection in iot networks using machine learning techniques: A review. *Asian journal of research in computer science*, 9(2):30–46.
- [Kairouz et al. 2021] Kairouz, P., McMahan, H. B., Avent, B., et al. (2021). Advances and open problems in federated learning. *Foundations and Trends in Machine Learning*, 14(1–2):1–210.
- [Karthikeyan et al. 2024] Karthikeyan, M., Manimegalai, D., and RajaGopal, K. (2024). Firefly algorithm based wsn-iot security enhancement with machine learning for intrusion detection. *Scientific Reports*, 14(1):231.
- [Kaushik et al. 2025] Kaushik, S., Bhardwaj, A., Almogren, A., Bharany, S., Altameem, A., Rehman, A. U., Hussien, S., and Hamam, H. (2025). Robust machine learning based intrusion detection system using simple statistical techniques in feature selection. *Scientific Reports*, 15(1):3970.

- [Khan et al. 2024] Khan, A. et al. (2024). Fedmade: Robust federated learning for intrusion detection in iot networks using a dynamic aggregation method. *arXiv preprint arXiv:2408.07152*.
- [Khraisat et al. 2021] Khraisat, A. et al. (2021). Intrusion detection in iot based on machine learning and deep learning algorithms. *Sensors*, 21(14):4751.
- [Li et al. 2020] Li, T., Sahu, A. K., Talwalkar, A., and Smith, V. (2020). Federated learning: Challenges, methods, and future directions. *IEEE Signal Processing Magazine*, 37(3):50–60.
- [Malathi and Padmaja 2023] Malathi, C. and Padmaja, I. N. (2023). Identification of cyber attacks using machine learning in smart iot networks. *Materials Today: Proceedings*, 80:2518–2523.
- [McMahan et al. 2017] McMahan, B., Moore, E., Ramage, D., Hampson, S., and Aguera y Arcas, B. (2017). Communication-efficient learning of deep networks from decentralized data. *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, pages 1273–1282.
- [Mishra et al. 2022] Mishra, S., Albarakati, A., and Sharma, S. K. (2022). Cyber threat intelligence for iot using machine learning. *Processes*, 10(12):2673.
- [Moustafa et al. 2020] Moustafa, N. et al. (2020). The ton_iot datasets: A new generation of datasets for iot intrusion detection systems. *arXiv preprint arXiv:2008.02846*.
- [Moustafa and Slay 2015] Moustafa, N. and Slay, J. (2015). Unsw-nb15 dataset.
- [Nguyen et al. 2025] Nguyen, M. et al. (2025). Federated learning for intrusion detection in iot environments: A privacy-preserving strategy. *Discover Internet of Things*.
- [Onsu et al. 2023] Onsu, M. A., Kantarci, B., and Boukerche, A. (2023). How to cope with malicious federated learning clients: an unsupervised learning-based approach. *Computer Networks*, 234:109938.
- [Rahman et al. 2025] Rahman, M. M., Shakil, S. A., and Mustakim, M. R. (2025). A survey on intrusion detection system in iot networks. *Cyber Security and Applications*, 3:100082.
- [S. and Nagapadma 2023] S., B. and Nagapadma, R. (2023). RT-IoT2022 . UCI Machine Learning Repository. DOI: <https://doi.org/10.24432/C5P338>.
- [Sanju 2023] Sanju, P. (2023). Enhancing intrusion detection in iot systems: A hybrid metaheuristics-deep learning approach with ensemble of recurrent neural networks. *Journal of Engineering Research*, 11(4):356–361.
- [Sharma et al. 2025] Sharma, P. et al. (2025). Edge-flguard: A federated learning framework for real-time anomaly detection in 5g-enabled iot ecosystems. *Applied Sciences*, 15(12):6452.
- [Tavallae et al. 2009] Tavallae, M. et al. (2009). Nsl-kdd dataset.
- [Tien et al. 2021] Tien, C.-W., Huang, T.-Y., Chen, P.-C., and Wang, J.-H. (2021). Using autoencoders for anomaly detection and transfer learning in iot. *Computers*, 10(7):88.

- [Vargas et al. 2021] Vargas, H., Lozano-Garzon, C., Montoya, G. A., and Donoso, Y. (2021). Detection of security attacks in industrial iot networks: A blockchain and machine learning approach. *Electronics*, 10(21):2662.
- [Zhang et al. 2024] Zhang, W. et al. (2024). Fedmse: Semi-supervised federated learning approach for iot network intrusion detection. *arXiv preprint arXiv:2410.14121*.
- [Zhang and Liu 2022] Zhang, Y. and Liu, Q. (2022). On iot intrusion detection based on data augmentation for enhancing learning on unbalanced samples. *Future Generation Computer Systems*, 133:213–227.